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Deficient TRPM3-linked mitochondrial Ca^{2+} influx in natural killer cells associated with myalgic encephalomyelitis/chronic fatigue syndrome

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Abstract

Introduction Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a multisystemic illness, commonly associated with dysregulation of the immune system including reduced cytotoxicity of natural killer (NK) cells and post-exertional neuroimmune exhaustion. Previously, transient receptor potential melastatin 3 (TRPM3) ion channel impairment associated with reduced Ca^{2+} mobilisation in NK cells from ME/CFS patients was reported. To further explore the pathomechanisms involved in ME/CFS, we investigated the downstream impact of TRPM3 ion channel dysfunction on mitochondrial Ca^{2+} mobilisation in NK cells.

Method Fluorescence live-cell imaging was used to investigate Ca^{2+} mobilisation in NK cells of ($N=10$) ME/CFS, classified using Canadian Consensus Criteria, and ($N=10$) healthy control (HC) participants. Cytoplasmic and mitochondrial Ca^{2+} entry was determined using Fluo-8 AM and Rhod-2 AM Ca^{2+} indicators, respectively. The effect of TRPM3 modulation on Ca^{2+} mobilisation ex vivo, was examined using pregnenolone sulfate and ononetin to activate and inhibit the channel, respectively.

Results Cytosolic Ca^{2+} influx amplitude and slope were significantly reduced ($p < 0.001$), with a significantly shorter $T_{1/2}$ response ($p = 0.001$) in ME/CFS compared to HC. Ca^{2+} influx amplitude ($p < 0.001$) and slope ($p < 0.041$) into the mitochondria were significantly higher in ME/CFS compared to HC. TRPM3 activation triggered pronounced cytosolic response ($P < 0.001$) accompanied by mitochondrial Ca^{2+} increase in HC. TRPM3-dependent cytosolic and mitochondrial Ca^{2+} mobilisation ($P < 0.015$) were significantly reduced with a shorter $T_{1/2}$ response ($p < 0.02$) in ME/CFS compared to HC.

Conclusion The results demonstrate that altered TRPM3-mediated cytosolic Ca^{2+} influx may significantly impact Ca^{2+} mobilisation into the mitochondria of people with ME/CFS. Alterations that interfere with the optimal function of Ca^{2+} permeable channels may cumulatively impact downstream signalling, leading to detrimental cellular consequences. Collectively these findings provide an avenue for further studies on the physiological functions of TRPM3 ion channel and its role in ME/CFS.

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Keywords TRP channels, TRPM3, Calcium signalling, Mitochondria, Myalgic encephalomyelitis, Chronic fatigue syndrome, ME/CFS, Natural killer cells

Background

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a complex and debilitating condition hallmarked by post-exertional malaise (PEM) and moderate to severe fatigue that cannot be relieved by rest [1]. ME/CFS affects multiple body systems and organs resulting in patients exhibiting a diverse range of symptoms predominantly associated with dysregulation of neurological, immunological, autonomic and endocrine body systems [1, 2]. Due to the absence of standardised diagnostic markers, diagnosis for ME/CFS is reliant on fulfilment of clinical case criteria, including the Canadian Consensus Criteria, 2003 (CCC) and the International Consensus Criteria, 2011 (ICC) [1, 2]. Despite affecting approximately 24 million people worldwide, ME/CFS aetiology remains elusive, in part due to the heterogeneous presentation of symptoms across multiple body systems [3].

Studies over the last few years have revealed a significant overlap in the clinical presentation of symptoms between ME/CFS and post-COVID-19 condition (PCC), a long-term sequelae of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection [4–7]. Both conditions are reported to have a profound negative impact on quality of life (QoL), causing significant financial strain to the healthcare system and those people directly affected due to ongoing medical needs [7–9].

Although ME/CFS is described as a multifactorial condition, dysregulation of the immune system with significant reduction in cytotoxicity by NK cells is reported as a consistent immunological finding [10]. The cytotoxic activity of primary NK cells and their target killing potential heavily rely on the precise regulation of intracellular Ca^{2+} ions for lytic granule polarisation, immune synapse formation and exocytosis of cytolytic proteins [11–13]. Therefore, maintenance of stable Ca^{2+} homeostasis conditions is crucial in NK cells, which is reliant on the synergistic network and function of receptors and organelles such the endoplasmic/sarcoplasmic reticulum (ER/SR) and the mitochondria [14–16]. In addition to their primary role as adenosine triphosphate (ATP) producers, the mitochondria possess the ability to buffer sustained cytosolic Ca^{2+} influx generated by activation of the SOCE process during immune cell activation [11, 17, 18]. This mitochondrial buffering action prevents excessive cytosolic Ca^{2+} rise, at the same time indirectly prolonging activation of Ca^{2+} ion permeable channels, shaping intracellular signals which is crucial for immune cell activation [19, 20].

Interferences to the Ca^{2+} mobilisation process due to dysregulation of Ca^{2+} permeable ion channels can

negatively impact intracellular signalling pathways [21]. Reduced NK cell cytotoxic activities associated with reduced Ca^{2+} mobilisation have been linked to significantly impaired transient receptor potential melastatin 3 (TRPM3) ion channel function in ME/CFS [22–24]. As a result, ME/CFS is described as a potential transient receptor potential (TRP) channelopathy. TRP channels have been described as important cellular sensors involved in numerous critical physiological processes, including temperature sensing, taste, vision, nociception, epithelial ion transport, and mineral homeostasis [25–27]. Therefore, dysregulation of expression and activity of TRP channels has been shown to be associated with some disease onset such as neuropathic pain, neurodegenerative diseases, and cancer, and collectively referred to as TRP-related channelopathies [28–30].

Due to relatively high permeability for Ca^{2+} , the TRPM3 channel play an important role in Ca^{2+} signalling, by mediating Ca^{2+} entry into the cells [26, 31]. Furthermore, TRPM3 channel activation was reported to induce pronounced downstream mitochondrial Ca^{2+} buffering in nociceptive dorsal root ganglion (DRG) neurons, modulating intracellular Ca^{2+} homeostasis and shaping intracellular Ca^{2+} -dependent signalling cascades [32, 33]. The significant reduction in TRPM3 surface protein expression [24] in conjunction with a significant reduction in Ca^{2+} influx in NK cells from ME/CFS patients [23] suggests important functional implications. Given TRP channels, pumps and the mitochondria are functionally coupled, and the activation pathways resulting in NK cell cytotoxicity is reliant on long-term sustained Ca^{2+} influx [11, 34], investigating the downstream effect of TRPM3 ion channel impairment on calcium mobilisation in ME/CFS was the focus of this investigation.

Materials and methods

Study participants

This research was conducted in accordance with Griffith University Human Research Ethics guidelines, the National Statement on Ethical Conduct in Human Research (2007, updated 2018), the Australian Code for the Responsible Conduct of Research (2018) and in accordance with Declaration of Helsinki. The human research ethics application was approved by the Griffith University Human Research Ethics Committee (GU HREC 2022/666) and all participants provided informed and written consent prior to blood collection. Clinical referral and the National Centre for Neuroimmunology and Emerging Diseases (NCNED) patient database were used for the recruitment of participants. Ten ME/

CFS patients and ten age and sex matched healthy controls (HC) volunteered for this study between 22 May 2024 and 20 March 2025. The inclusion criteria required ME/CFS patients to have received a clinical diagnosis and meeting the ICC (2011) and CCC (2003) clinical case definitions. HC participants were included if in good health without incidence of fatigue or evidence of illness. The exclusion criteria applied removed participants if they were pregnant or breastfeeding, had documented history of alcohol abuse, had body mass index (BMI) ≥ 30 , younger than 18 years or over 65 years, or presented medical conditions. These conditions included chronic fatigue, autoimmune diseases, metabolic syndrome, cardiovascular disease, diabetes, thyroid disease, malignancies, insomnia, and primary psychological disorders. Participants who were regularly using medications that directly or indirectly influence TRPM3 or Ca^{2+} signalling were provided with the option to cease for a minimum of 14 days prior to blood donation after approval from their physician.

Sociodemographic background, medical history, medications taken, symptom presentation for cases, and quality of life (QoL) information were acquired from participants using online questionnaires. Validated QoL measures included the 36-item short form health survey (SF-36) and the WHO Disability Assessment Schedule (DAS). The SF-36 was evaluated across eight domains: physical functioning, role limitations due to physical health problems, bodily pain, overall health perceptions, energy and vitality, social functioning, role limitations due to personal or emotional problems, and emotional wellbeing or general mental health. A 0% to 100% scale was used to evaluate QoL, whereby 0 indicated very severe and 100 being no symptom. The disability level was evaluated using the WHODAS form across six domains of life: cognition, mobility, self-care, getting along, work and school participation, general life activities, and participation in society (Available online at: <https://apps.who.int/iris/handle/10665/43974>.) [35]. Low percentage scores were associated with no disability while a high degree of difficulty or greater disability was represented by higher percentages [36]. Symptoms were categorised based on presence, frequency, and severity in 10 different groups including cognitive difficulties; pain; sleep disturbances; cardiovascular, gastrointestinal, respiratory, urinary, immunological, and thermostatic intolerances; and sensory disturbances.

Natural killer cells isolation

Blood collections were performed between 6:30 am and 10:00 am in patient homes, around the Gold Coast campus of Griffith University, Robina Hospital and at private laboratories in South East Queensland and North-East New South Wales. Venepuncture was used to collect

84 ml of whole blood in ethylenediaminetetraacetic acid (EDTA) tubes. Full blood counts for all participants were conducted at National Association of Testing Authorities (NATA)-accredited facilities, including Gold Coast University Hospital or private laboratories measuring red blood cells, white blood cells and granulocyte cells within 4 h of collection.

Prior to arriving at the NCNED laboratories, all samples were de-identified using a unique alphanumeric code. PBMCs were isolated from 80 ml of whole blood samples following our previously described protocol [37], using density gradient centrifugation [Ficoll Paque Plus (Cytiva), GE Healthcare, Uppsala, Sweden]. Trypan blue stain (Invitrogen, Carlsbad, CA, USA) and an automatic cell counter (TC20 Automated cell counter, Bio-Rad, Laboratories, Hercules, CA) were used for cell count and cell viability determination. The EasySep human NK cell enrichment kit (Stem Cell Technologies, Vancouver, BC, Canada) was used for NK cell isolation from PBMCs by immunomagnetic selection following our previous protocol [37]. For purity analysis, NK cells were incubated for 20 min at room temperature (RT) in the presence of CD3 PE-Cy7 (5 μl /test) and CD56 APC (20 μl /test) monoclonal antibodies (Becton Dickinson (BD) Biosciences, San Diego, CA, USA). NK cells were identified as $\text{CD3}^{-}\text{CD56}^{+}$ populations, acquired at 10,000 events using the BD X20 flow cytometer (BD Biosciences, San Diego, CA, USA). Acceptable NK cell purity was $\geq 75\%$ (Supplementary Fig. 1).

Cell staining

EDTA and Ca^{2+} free buffer was prepared in milliQ water [0 mM Ca^{2+} buffer: Sodium chloride (NaCl) 140 mM, Potassium chloride (KCl) 5.4 mM, Magnesium chloride (MgCl_2) 2.1 mM, and HEPES 10 mM]. 1.8 mM Ca^{2+} buffer was prepared in milliQ water [1.8 mM Ca^{2+} buffer: NaCl 140 mM, KCl 5.4 mM, Calcium chloride (CaCl_2) 1.8 mM, MgCl_2 1.0 mM, and HEPES 10 mM]. For both buffers, the pH was adjusted to 7.40 ± 0.05 using NaOH and the osmolality to 300 ± 10 mOsm/L using D-glucose. Stock solutions for PregS, Ononetin, Thapsigargin, Fluo-8 AM, Mitotracker Green, Rhod-2 were prepared in Dimethylsulfoxide (DMSO) (Sigma-Aldrich).

NK cells were immobilised in 24-well glass bottom plates (Celvis, P24-1.5 H-N) using Corning® Cell-Tak™ Cell and Tissue Adhesive (Bio-Strategy, Campbellfield, Vic, Australia) as previously described [38]. NK cells were washed and stained in 0 mM Ca^{2+} buffer unless stated. Cells were stained with 1 μM Fluo-8-acetoxymethyl ester (AM) (Abcam, Cambridge, UK) in the presence of 0.02% Pluronic F127 for 30 min at 37 °C, then washed. To improve localisation of Rhod-2 into the mitochondria, a small solid excess of sodium borohydride (NaBH_4) was added to Rhod-2, reducing the dye to a colourless

nonfluorescent dihydropyrene-2 AM. For mitochondria specific calcium mobilisation determination, NK cells were first stained with 1 μ M Mito Tracker Green FM for 30 min at 37 °C, washed, then stained with 1 μ M Hoechst 33,342 and 1 μ M Rhod-2 AM in the presence of 0.02% Pluronic F127 (ThermoFisher, Waltham, MA, USA) for 15 min at RT. Protocol optimisations were performed to confirm depletion of ER Ca^{2+} stores, and all cells were pre-depleted with 1 μ M Thapsigargin (Sigma-Aldrich) for 15 min at RT, then left for a further 15 min allowing de-esterification of Fuo-8-AM and Rhod-2-AM, ready for imaging.

For TRPM3 modulation, a continuous gravity perfusion system was used to administer PregS and Ononetin in 1.8 mM Ca^{2+} buffer. The Evident IXplore IX83 SpinSR Super Resolution microscope system (60x objective, oil immersion, spinning disk confocal configuration) with a Hamamatsu ORCA-Fusion camera and Olympus Cellsens Dimension 4.2 acquisition software (Evident Corporation, Tokyo, Japan) was used for fluorescence live cell imaging of cells in 24-well glass bottom plates [Fluo-8, Excitation/Emission (Ex/Em) 489/508 nm; Mito Tracker Green FM, Ex/Em: 490/516 nm; Rhod-2, Ex/Em: 552/581 nm and Hoechst 33342, Ex/Em: 341/542 nm]. All live cell immunofluorescent imaging was performed at RT (20–25 °C) to ensure non-specific organelle uptake of Ca^{2+} indicators did not occur. Furthermore, experiments performed at RT enabled replication of drug induced ion channel response condition previously reported with electrophysiology findings in an alternative method and reduced the potential impact confounding data due to thermoregulatory activation at physiological temperatures. Time lapse images were acquired at 15s intervals. TRPM3-dependent Ca^{2+} mobilisation was assessed using live cell imaging employing the following sequence: 60s of baseline imaging in 0 mM Ca^{2+} ; addition of 1.8mM Ca^{2+} for 3 min; TRPM3 activation with 50 μ M PregS (In vitro technology) for 3 min; finally, 1 μ M Carbonyl cyanide 4-(trifluoromethoxy)phenylhydrazone (FCCP) for 3 min to monitor and indicate mitochondrial uncoupling. 0 mM Ca^{2+} solution was used to wash the perfusion line between subsequent recordings. To confirm TRPM3-dependent Ca^{2+} mobilisation, 4 μ M Ononetin was applied to inhibit TRPM3 employing the following sequence: 60s of baseline imaging in 0 mM Ca^{2+} ; addition of 1.8 mM Ca^{2+} buffer for 3 min; inhibition by 4 μ M Ononetin alone for 3 min; combined treatment with 50 μ M PregS and 4 μ M Ononetin for 3 min; TRPM3 activation by PregS alone for 3 min; 1 μ M FCCP to induce mitochondrial uncoupling for 3 min (S3 Fig).

Image analysis

Olympus cellSens Dimension Desktop 4.3 software was used for time lapse image data extraction, selecting

regions of interest (ROIs) per well. A ROI was selected on an area without visible cells for background subtraction. Cells with an intact membrane and visible throughout the recording duration were selected. Mitochondrial networks were identified by drawing ROIs around Mito Tracker Green perimeter and Rhod-2 colocalization. Colocalization of the two stains was measured using ImageJ [JACoP (Just Another Colocalization Plugin) software [39], employing Pearson's correlation coefficient (PCC) and Manders' Colocalization Coefficients (MCC). M1 = red, fraction of Rhod-2 overlapping green (Mito-tracker Green); M2 = fraction of Mitotracker Green overlapping red (Rhod-2). Fluorescence intensity raw data output was analysed using programmable excel spreadsheets [Baseline fluorescence: 0–60s in 0 mM Ca^{2+} solution; F0: Minimum fluorescence; Fmax: Fluorescence at maximum peak; T1: Time at minimum fluorescence intensity; T2: Time at maximum fluorescence intensity; Relative Fluorescence = $\text{Fmax}/\text{F0}$; Slope = $[(\text{Fmax}-\text{F0})/(\text{T2}-\text{T1})]$; Half-time ($\text{T}_{1/2}$) of maximal response = $\text{F}_{1/2\text{max}}$].

Biological and chemical reagents

NaHCO_3 (product: S5761-500 g), NaCl (product code: 793566-500G), KCl (product code: 793590-500G), CaCl_2 (product code: C5670-100G), MgCl_2 (product code: 63069-100ML), NaBH_4 (product code: 452882–100 g), FCCP (product code: C2920), Thapsigargin (product code: T9033-1MG), DMSO (product code: D2650-5 $\times$ 10 ml) were purchased from Sigma-Aldrich. RPMI 1640 (product code: 11835030), HEPES (product code: 11835030), Mito Tracker TM Green FM (M7514), Rhod 2 AM (R1245MP), Hoechst 33,342 (H3570) and Pluronic F-127 (P3000MP) were purchased from Thermo Fisher Scientific. EasySep human NK cell enrichment kits (product code: 19055) and EasySep buffer (product code: 20144) were purchased from Stem Cell Technologies. Ficoll paque plus (product code: GEHE17-1440-03) and Cell-Tak cell adhesion (product code: BDAA354240, Bio-Strategy) were purchased from Bio-Strategy, Australia. 24 Well glass bottom plates with high performance #1.5 cover glass (product code: P24-1.5 H-N) were purchased from Cellvis, USA. PregS (product code: RDS537650) and Ononetin (product code: RDS514350) were purchased from In Vitro Technologies.

Statistical analysis

Data and statistical analyses were performed using Microsoft Excel Version 2510, Origin software version 2024b (OriginLab Corporation Massachusetts, USA), Statistical Package for the Social Sciences (SPSS) v26 (IBM Corp, Version 29, Armonk, NY, USA) and Graph-Pad Prism, version 9 (GraphPad software Inc., CA, USA). Descriptive statistics were used to analyse the

frequencies and percentages for participant characteristics and responses. Outliers were identified and removed using the Robust regression and Outlier removal (ROUT) method. GraphPad formed statistical comparison by participant, not by individual cell, eliminating risk of pseudoreplication [40]. Cell recordings were excluded if they returned an outlier value for one of the parameters (slope, $T_{1/2}$, and amplitude) to minimise variability. The Shapiro-Wilk test was used for normality test, with the Mann Whitney U tests performed on nonparametric data. A p -value < 0.05 was considered statistically significant.

Results

Participants demographics

Twenty participants were included in this study, $N = 10$, HC and $N = 10$, ME/CFS patients who met the ICC or CCC clinical case definitions [1, 2] and did not report other fatigue related illnesses that explain their symptoms. Demographic data are presented in Table 1. For ME/CFS patients, the average age and SD were $43.40 \pm$

13.72 years and 44.90 ± 11.09 years for HC. Eight HC and eight ME/CFS were female participants. Two HC and two ME/CFS were male participants. There was no significant difference in gender, age and BMI among groups.

WHO DAS and SF-36 surveys were used to assess the QoL and disability for all participants. Data is shown in Table 1. There was significant difference identified between HC and ME/CFS patients in all domains of SF-36 including physical functioning, physical role, social functioning and vitality ($p=0.001$), general health ($p=0.001$), pain ($p=0.001$), emotional wellbeing ($p=0.002$), and emotional role ($p=0.004$). WHO DAS scores were significantly different between the two groups in all domains. Communication and understanding ($p=0.007$), mobility ($p=0.001$), life activities ($p=0.001$), self-care ($p=0.007$), interpersonal relationships ($p=0.007$), work/school participation ($p=0.001$), and participation in society ($p=0.001$).

Participant demographics quality of life and disability scores (SF-36 and WHODAS)

Table 1 Variations in QoL and disability scores between HC and ME/CFS patients, based on SF-36 and WHODAS surveys

	HC	ME/CFS	P-value
Age (years)	44.90±11.09	43.40±13.72	0.791
Gender Male\Female	2/8	2/8	
BMI (kg/m ²)	22.93±2.86	23.82±3.19	0.519
SF-36 (%)			
Physical Functioning	97.78±5.07	26.67±12.99	0.001
Physical Role	95.83±10.36	5.56±6.59	0.001
Pain	88.33±13.64	35.83±16.96	0.001
General Health	74.07±15.41	22.69±10.44	0.001
Social Functioning	90.28±17.43	27.78±16.27	0.001
Emotional Role	100.00±0.00	56.48±28.80	0.002
Emotional Wellbeing	80.56±15.50	58.89±11.67	0.004
Vitality	65.97±23.20	8.33±7.65	0.001
WHODAS (%)			
Communication & Understanding	16.66±28.94	51.85±12.17	0.007
Mobility	0.00±0.00	60.56±11.84	0.001
Self-care	0.00±0.00	27.78±23.20	0.007
Interpersonal Relationships	6.25±7.65	38.21±26.79	0.007
Life Activities	8.33±14.99	63.89±20.68	0.001
Work/School Participation	2.78±6.34	60.18±32.16	0.001
Participation in Society	3.01±4.43	56.80±20.32	0.001

A 0% to 100% scale was used to evaluate QoL whereby high SF-36 scores reflected good QoL, and low scores indicated poor QoL (0=severe symptom, 100=no symptom)

Low WHODAS scores were associated with no functional limitations and high scores reflected severe functional impairments (0=no symptom, 100=very severe)

Missing data from one HC participant who did not complete the WHODAS and SF-36 surveys. Data is presented as mean±SD or percentage (%)

Abbreviations BMI body mass index, HC healthy controls, ME/CFS myalgic encephalomyelitis / chronic fatigue syndrome, SF-36 36-item short-form health survey, WHODAS World Health Organization disability assessment schedule

Values of $p < 0.05$ are bolded

Symptoms and onset details from ME/CFS Group

The average age and SD of illness onset for ME/CFS patients was 27.00 ± 12.55 years, with 16.70 ± 11.00 years as the average and SD for disease duration. All 10 (100.0%) ME/CFS patients reported experiencing fatigue, sleep disturbances, cognitive difficulties, sensory disturbances and pain. Gastrointestinal and cardiovascular disturbances were declared by 7 (70.0%), while immune disturbances by 6 (60.0%). Respiratory problems were reported by 5 (50.0%), thermotactic intolerance by 2 (20.0%) and urinary disturbances by 4 (40.0%). All symptoms and frequency data are detailed in Table 2.

Full blood count parameters

A full blood count was performed for all participants. The results were within the normal range for all cases as defined by pathology service providers. Detailed results in Table 3.

Reduced TRPM3-dependent Cytosolic Ca²⁺ Mobilisation in NK Cells from ME/CFS Patients

Fluorescence live cell imaging data acquired by measuring the relative amount of Ca²⁺ influx and rate of response using Fluo-8 AM calcium indicators are presented in Fig. 1. Cytosolic Ca²⁺ influx data is presented as the relative change in fluorescence intensity of Fluo-8 AM in Fig. 1. Confirming previous studies, the rate of response (Fig. 1D, $p < 0.001$) and average Ca²⁺ influx amplitude (Fig. 1B, $p < 0.001$) into the cytosol was significantly reduced in ME/CFS patients compared to HC, following the addition of 1.8mM Ca²⁺ buffer. Further, the time to half-maximal ($T_{1/2}$) response was significantly

Table 2 The symptoms and onset details from ME/CFS group

		ME/CFS
Age of onset (Years [Mean $\pm$ SD])		27.00 $\pm$ 12.55
Disease duration (Years [Mean $\pm$ SD])		16.70 $\pm$ 11.00
Fatigue	Yes	10 (100.0%)
	No	0 (0.0%)
Cognitive Difficulties	Yes	10 (100.0%)
	No	0 (0.0%)
Pain	Yes	10 (100.0%)
	No	0 (0.0%)
Sleep Disturbances	Yes	10 (100.0%)
	No	0 (0.0%)
Sensory Disturbances	Yes	10 (100.0%)
	No	0 (0.0%)
Immune Disturbances	Yes	6 (60.0%)
	No	4 (40.0%)
Gastrointestinal Disturbances	Yes	7 (70.0%)
	No	3 (30.0%)
Cardiovascular Disturbances	Yes	7 (70.0%)
	No	3 (30.0%)
Respiratory	Yes	5 (50.0%)
	No	5 (50.0%)
Thermostatic Intolerance	Yes	2 (20.0%)
	No	8 (80.0%)
Urinary Disturbances	Yes	4 (40.0%)
	No	6 (60%)

Data is presented as mean $\pm$ SD or (%)

Abbreviations ME/CFS myalgic encephalomyelitis / chronic fatigue syndrome, SD standard deviation, N number of participants

Table 3 Parameters for full blood count between groups

		HC	ME/CFS	P-value
Full Blood Count	White Cell Count (4.0–11.0 $\times 10^9$ /L)	5.26 $\pm$ 1.17	5.52 $\pm$ 1.35	0.652
	Lymphocytes (1.0–4.0 $\times 10^9$ /L)	1.67 $\pm$ 0.32	1.80 $\pm$ 0.60	0.581
	Neutrophils (2.0–8.0 $\times 10^9$ /L)	3.09 $\pm$ 0.97	3.07 $\pm$ 1.04	0.967
	Monocytes (0.1–1.0 $\times 10^9$ /L)	0.36 $\pm$ 0.08	0.41 $\pm$ 0.07	0.161
	Eosinophils (< 0.6 $\times 10^9$ /L)	0.08 $\pm$ 0.04	0.17 $\pm$ 0.12	0.049
	Basophils (< 0.2 $\times 10^9$ /L)	0.03 $\pm$ 0.01	0.06 $\pm$ 0.04	0.070
	Platelets (140–400 $\times 10^9$ /L)	246.50 $\pm$ 59.77	254.90 $\pm$ 48.30	0.734
	Red Cell Count (3.8–5.2 $\times 10^{12}$ /L)	4.48 $\pm$ 0.25	4.59 $\pm$ 0.55	0.573
	Haematocrit (0.33–0.47)	0.40 $\pm$ 0.02	0.42 $\pm$ 0.04	0.783
	Haemoglobin (115–160 g/L)	134.20 $\pm$ 8.31	135.60 $\pm$ 13.50	0.395

Data are presented as mean $\pm$ SD

Reference ranges for full blood count parameters are provided in the table

Abbreviations FBC Full Blood Count, HC healthy controls, ME/CFS myalgic encephalomyelitis / chronic fatigue syndrome, N number of participants

shorter in ME/CFS patients compared to HC (Fig. 1E, $p = 0.001$). Stimulation of the TRPM3 ion channel with 50 μ M PregS successfully triggered a significantly higher rate of response (Fig. 1D, slope, $P < 0.001$) in HC compared to ME/CFS patients. In contrast, PregS induced a significantly reduced rate of response (Fig. 1D, $p < 0.001$) and reduced cytosolic Ca^{2+} influx amplitude (Fig. 1c, $P < 0.015$), with a shorter $T_{1/2}$ response (Fig. 1E, $p = 0.02$) in ME/CFS compared to HC. Further, 4 μ M ononetin was used to inhibit TRPM3 activity, subsequently confirming TRPM3-dependent Ca^{2+} mobilisation following TRPM3 activation by 50 μ M PregS (Supplementary Fig. 3). Effective inhibition of TRPM3 ionic currents following simultaneous application of ononetin and PregS in NK cells from HC was previously shown using the patch-clamp technique [22, 41].

Reduced TRPM3-dependent Ca^{2+} Mobilisation into the Mitochondria in NK Cells from ME/CFS Patients

The downstream effect of TRPM3 ion channel impairment on Ca^{2+} influx into the mitochondria was examined using mitochondrial Ca^{2+} indicator Rhod-2. Despite a short $T_{1/2}$ response (Fig. 2E) in ME/CFS compared to HC, the rate of response (Fig. 2D, slope, $p = 0.041$) and mitochondrial Ca^{2+} influx amplitude (Fig. 2C, $p = 0.001$) were significantly increased. Contrary, PregS successfully induced a significantly higher TRPM3-dependent response rate (Fig. 2D, slope, $p = 0.001$) in HC. Thus, Ca^{2+} influx amplitude (Fig. 2C, $p = 0.001$) into the mitochondria was significantly reduced in ME/CFS compared to HC. Furthermore, the $T_{1/2}$ response (Fig. 2E, $p = 0.001$) for PregS was significantly shorter in ME/CFS patients. To monitor and confirm mitochondrial Ca^{2+} signalling downstream TRPM3-activation by PregS, 1 μ M FCCP was added to cells. Indeed, the signal was attenuated indicating mitochondrial uncoupling. Further, to demonstrate a potential downstream signalling of TRPM3-dependent Ca^{2+} influx into the mitochondria, NK cells stained with Fluo-8 and Rhod-2 were simultaneously imaged (Supplementary Fig. 2). We successfully detected TRPM3-dependent Ca^{2+} influx response and the downstream signalling into the mitochondria in HC, however the responses in ME/CFS were highly disturbed.

Discussion

Our previous studies in NK cells from patients with ME/CFS have reported impairment of the TRPM3 ion channel associated with reduced cytosolic Ca^{2+} mobilisation [22–24, 42]. The functional deficiency of NK cells and the potential link to the pathological mechanisms involved in ME/CFS has been widely reported [10, 43]. The collaborative function between plasma membrane Ca^{2+} permeable channels, ER Ca^{2+} sensors and the mitochondria which act as Ca^{2+} buffers, carefully coordinate the

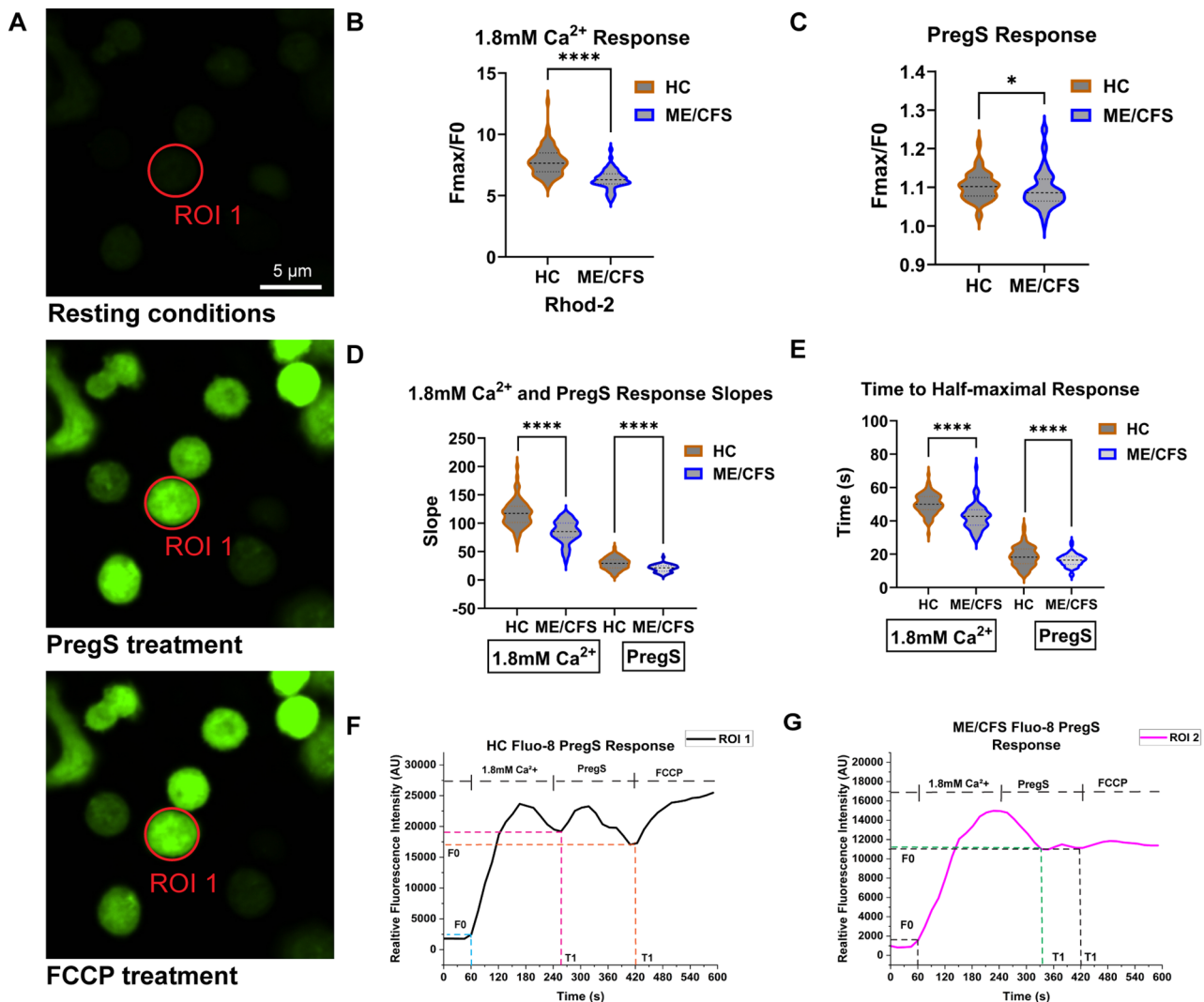


Fig. 1 Cytosolic Ca^{2+} influx response in freshly isolated NK cells. **A** Examples of NK cells from a HC stained with Fluo-8 intracellular calcium indicator imaged at 60 $\times$, at the start of the experiment, after stimulation with PregS and FCCP, respectively. Group comparisons for pooled data (**B–E**) were performed with unadjusted two-sided Mann-Whitney U tests. The violin plots show median (central line) and quartile levels for pooled data. * Represents significant differences after group comparison. **B** Pooled cytosolic Ca^{2+} influx in HC ($N = 10$, $n = 554$ NK cells) compared with ME/CFS ($N = 10$, $n = 415$ NK cells) following addition of 1.8mM Ca^{2+} to NK cells. Fluorescence intensity is shown as the ratio of measured intensity (Fmax) and the minimum fluorescence value (F0). **C** Response to 50 μM PregS stimulation in HC ($N = 10$, $n = 398$ NK cells) and ME/CFS ($N = 10$, $n = 281$ NK cells) (d) the rate (slope) of response to 1.8mM Ca^{2+} and PregS. **E** $T_{1/2}$ response for 1.8mM Ca^{2+} and PregS. **F** An example of Fluo-8 AM time lapse fluorescence intensity curve for one cell, ROI 1 shown in (**A**) for NK cells from a HC, **G** An example of time lapse fluorescence intensity curves for a single NK cell from an ME/CFS patient. The dotted lines indicate F0s and corresponding T1s for each stimulation condition (f: Light blue, baseline; magenta, PregS response; orange, FCCP response; G: black, baseline; green, PregS response and black, FCCP response). N =number of participants and n =number of cells analysed per group. Abbreviations: Ca^{2+} , calcium; HC, healthy control; ME/CFS, myalgic encephalomyelitis / chronic fatigue syndrome; ROI, region of interest; FCCP, Carbonyl cyanide 4-(trifluoromethoxy) phenylhydrazone; Fmax, maximum fluorescence; F0, minimum fluorescence; PregS, Pregnenolone sulphate. Imaging analysis was performed using cellSens Dimension Desktop, then edited using Adobe Illustrator. Time lapse fluorescence Graphs were constructed using Origin software

cytotoxic activity of NK cells [34, 44, 45]. These upstream signalling networks propagate activating intracellular signalling networks responsible for the effector functions of NK cells [46]. Hence, the current study focused on investigating how TRPM3 ion channel dysfunction impact downstream Ca^{2+} mobilisation into the mitochondria.

Results from this present study further supports the hypothesis of TRPM3 ion channel dysfunction in NK

cells from ME/CFS patients and the negative impact on Ca^{2+} mobilisation into cellular organelles, potentially contributing to the pathomechanisms involved in this condition. Downstream intracellular signalling cascades are shaped by the distribution and function of Ca^{2+} permeable channels which influence the function and activity of NK cells by regulating Ca^{2+} homeostasis [44]. Hence, we first confirmed our previous findings by

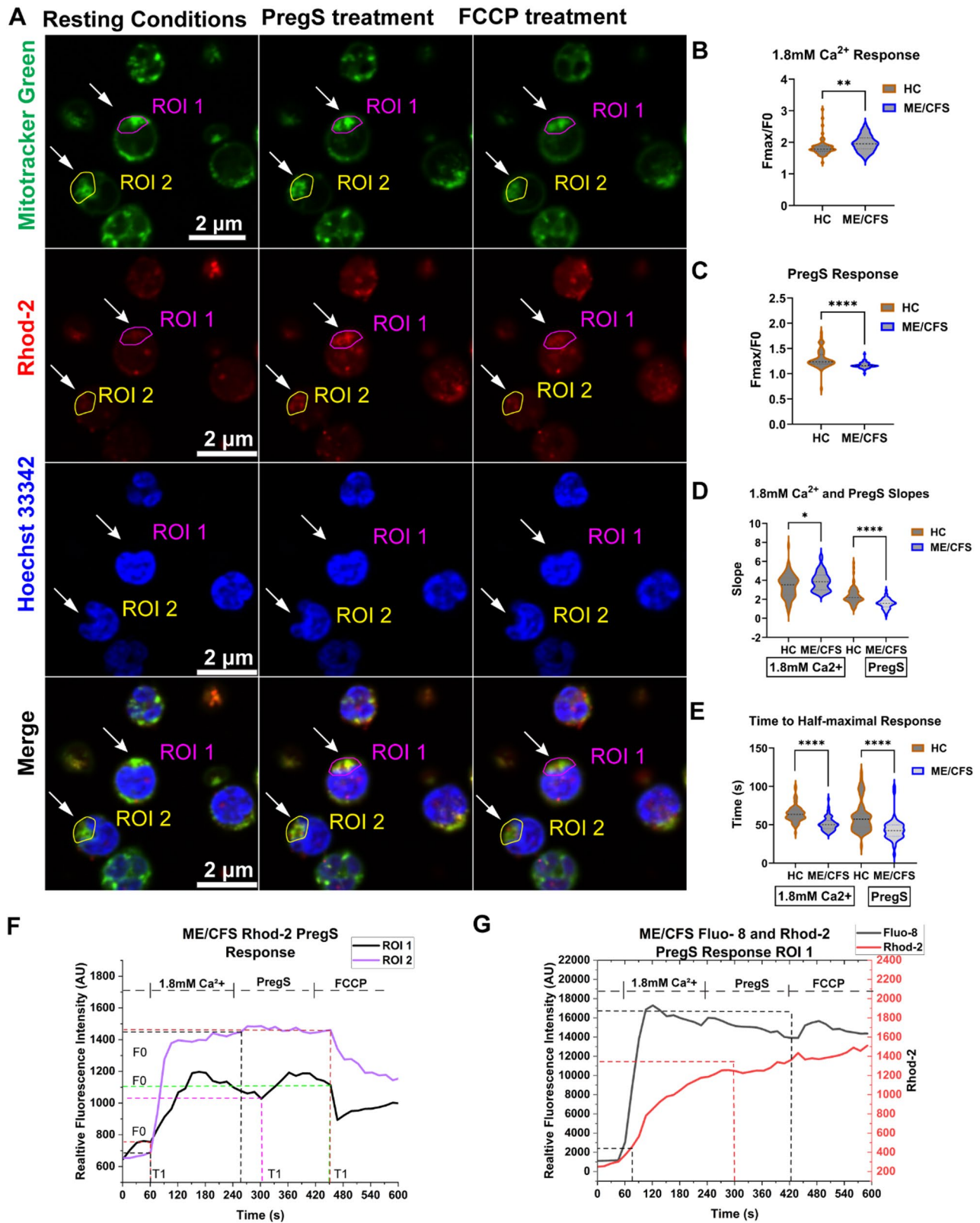


Fig. 2 (See legend on next page.)

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Fig. 2 Fluorescence imaging of calcium influx into the mitochondria in NK cells. **A** Representative images of NK cells from an ME/CFS patient stained with Mitotracker Green FM (green), Rhod-2 AM (red) and Hoechst 33342 (blue) taken at the start of the experiment, after PregS and FCCP stimulation, respectively at 60 x. Arrows indicate ROI perimeter around mitochondrial network, indicating change in Rhod-2 fluorescence intensity overtime (increased yellow spots overtime). Group comparisons for pooled data (**B–E**) were performed with unadjusted two-sided Mann-Whitney U tests. The violin plots show median (central line) and quartile levels for pooled data. * Represents significant differences after group comparison. **B & C** Comparison of responses to 1.8mM Ca^{2+} in ME/CFS ($N=10$, $n=372$ NK cells) and HC ($N=10$, $n=437$ NK cells) and PregS stimulation in ME/CFS ($N=10$, $n=349$ NK cells) and HC ($N=10$, $n=378$ NK cells), respectively. **D** Rate (slope) of responses, **E** $T_{1/2}$ response to 1.8mM Ca^{2+} and PregS, respectively. **F** Examples of Rhod-2 AM time lapse fluorescence intensity curves for two NK cells (ROI 1 and 2) from an ME/CFS patient shown in **(A)**. **G** An example of Rhod-2 AM time lapse fluorescence intensity curve for a single NK cell (ROI 1) from a HC. The dotted lines indicate F0s and corresponding T1s for each stimulation condition (f: black and orange, baseline; black and magenta, PregS responses and green and orange, FCCP responses; g: black, baseline; orange, PregS response and black, FCCP response). N=number of participants and n=number of cells analysed per group. Abbreviations: Ca^{2+} , calcium; HC, healthy control; ME/CFS, myalgic encephalomyelitis / chronic fatigue syndrome; ROI, region of interest; 4-(FCCP, Carbonyl cyanide trifluoromethoxy) phenylhydrazine; Fmax, maximum fluorescence; F0, minimum fluorescence; PregS, Pregnenolone sulphate. Imaging analysis was performed using cellSens Dimension Desktop, then edited using Adobe illustrator. Time lapse fluorescence Graphs were constructed using Origin software

comparing Ca^{2+} mobilisation in response to addition of 1.8mM Ca^{2+} solution, following depletion of internal Ca^{2+} stores using thapsigargin [23, 47]. Depletion of ER stores activates the SOCE process, evoking Ca^{2+} signalling and activation of additional Ca^{2+} permeable channels such as TRP channels, subsequently triggering Ca^{2+} mobilisation into the cytoplasm and replenishment of Ca^{2+} stores [44, 48, 49]. Importantly, the rate of response and Ca^{2+} influx amplitude into the cytosol was significantly reduced with a shorter $T_{1/2}$ response in ME/CFS patients compared to HC. Hence, demonstrating ion channel dysfunction and the subsequent cytosolic Ca^{2+} mobilisation disturbances in NK cells from ME/CFS patients as we previously reported [23].

Immune dysregulations such as a heightened immune system, high inflammation markers, disrupted immune pathways and reduced cytotoxicity by NK cells have been reported in ME/CFS patients [10, 23, 50–52]. NK cells have been reported to undergo phenotypic, regulatory and functional changes during viral infections, significantly impacting their functional activity [53]. In ME/CFS, ion channel dysfunction with reduced intracellular Ca^{2+} influx following phosphorylation of the mitogen-activated protein kinases (MAPK) pathway in NK cells has also been reported [54]. Phosphorylation of MAPK subgroups, mitogen-activated protein kinase 1/2 (MEK1/2) and p38 regulate cytokine production, and phosphorylation of extracellular signal-regulated kinases (ERK)1/2 polarises the secretory granule towards immune synapse [55]. Hence, considering NK cell target killing functions are governed by carefully regulated Ca^{2+} flux, dysregulation of TRP ion channels has detrimental consequences. Given the distribution and functional states of Ca^{2+} permeable channels, fundamentally impact Ca^{2+} mobilisation across cells [14–16, 21], we investigated TRPM3-dependent Ca^{2+} mobilisation into the cytosol. Stimulation of the TRPM3 ion channel with 50 μM PregS successfully triggered a higher rate of response and cytosolic Ca^{2+} influx in HC ($P < 0.001$). Conversely, the cytosolic rate of response and Ca^{2+} influx amplitude ($P < 0.001$) were significantly reduced with a shorter

$T_{1/2}$ response in ME/CFS compared to HC. These findings indicate and confirm that TRPM3 is functionally impaired and the duration for channel activity is diminished in ME/CFS. Alterations that interfere with the optimal function of Ca^{2+} channels may cumulatively impact downstream signalling cascades, leading to detrimental cellular consequences, in this instance decrement in Ca^{2+} mobilisation in NK cells, potentially contributing to reduced cytotoxicity properties and other disturbances reported in ME/CFS. Additionally, as TRPM3 expression is ubiquitously expressed in tissue, organs and cells such as the brain, spinal cord, kidney and pancreatic beta cells [25, 26], the impaired TRPM3-dependent Ca^{2+} mobilisation may potentially be present in other tissues, contributing to sensory disturbances, cognitive dysfunction, chronic fatigue and other disturbances exhibited in ME/CFS [56].

The complex interplay between plasma membrane Ca^{2+} permeable channels, ER Ca^{2+} sensors and the mitochondria, carefully coordinate mechanism of Ca^{2+} influx in NK cells [14, 45, 57]. Importantly, TRPM3 activation was found to trigger mitochondrial Ca^{2+} loading in DRG neurons, highlighting dynamic mitochondrial activity downstream of TRPM3 activation [32]. Given the functional coupling and dynamic interplay exhibited by these organelles, it was deemed relevant to investigate TRPM3-dependent Ca^{2+} influx into the mitochondria in NK cells from ME/CFS patients. Despite a short $T_{1/2}$ response (Fig. 3E) in ME/CFS compared to HC, the rate of response (Fig. 3D, slope, $p = 0.041$) and mitochondrial Ca^{2+} influx amplitude (Fig. 3C, $p = 0.001$) were significantly increased.

Considering that Ca^{2+} influx into the cytoplasm was significantly reduced via TRPM3 stimulation in NK cells from ME/CFS patients, we theorise that Ca^{2+} storage-amount in the mitochondria is potentially reduced, consequently causing rapid (increased slope) Ca^{2+} uptake to replenish Ca^{2+} stores. In addition, it is also possible that interferences due to dysfunctional TRP channels trigger altered Ca^{2+} signalling processes downstream. Alterations in the Ca^{2+} flux pathways and speed, were found

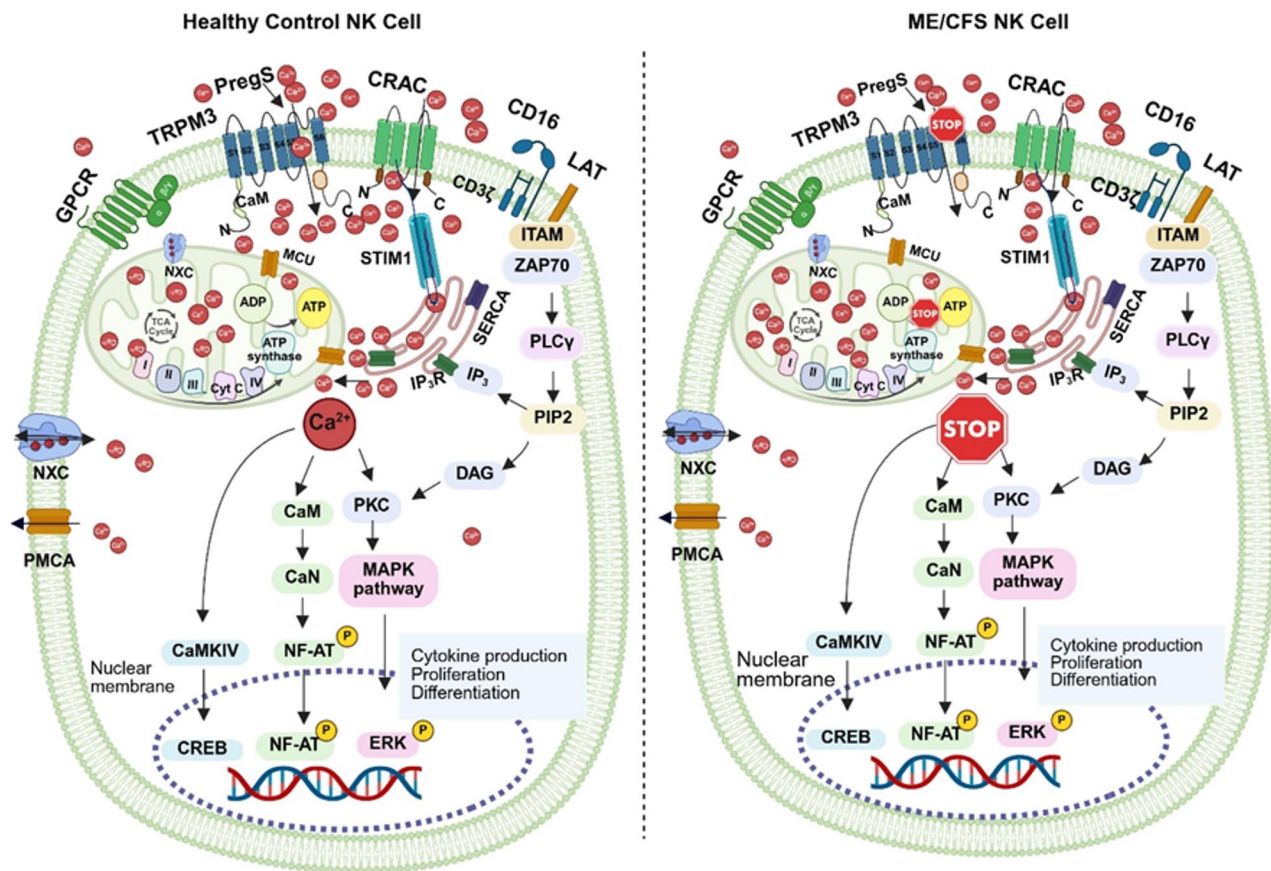


Fig. 3 A schematic illustration of the proposed conceptual NK cell models in HC and ME/CFS. Ligand binding or TRPM3 stimulation induces a rise in cytosolic and mitochondrial Ca^{2+} levels. NK cell receptor engagement activates PLC- γ , initiating a two-stage intracellular Ca^{2+} rise; Ca^{2+} release from ER stores and SOCE through plasma membrane channels including TRPM3. Activated PLC- γ hydrolyses PIP_2 into IP_3 and DAG. IP_3 binds to IP_3 receptors and drives Ca^{2+} release from ER into the cytoplasm and mitochondria. STIM1 senses ER Ca^{2+} depletion, translocate to towards plasma membrane and interacts with Orai1, leading to activation of CRAC channels and Ca^{2+} influx via several Ca^{2+} permeable channels including TRPM3. Increased cytosolic Ca^{2+} levels leads to activation of downstream signalling events including ATP production, MAP kinases (Raf, ERK1/2, JNK) and transcription factors (CREB, AP-1 and NF-AT), regulating proliferation, differentiation and immune cell activation (the expression for genes encoding potent immunomodulatory cytokines, including IL-2, IL-3, IL-4, TNF- α , and GM-CSF). In NK cells, impaired TRPM3 channel function may lead to dysregulated Ca^{2+} signalling, subsequently disrupting ATP production, and Ca^{2+} -dependent signalling events, thereby impacting cytotoxic activity and broader immune dysfunction. Figure 3 highlight the proposed potential consequences of TRPM3 impairment in ME/CFS patients. Abbreviations: CaM, calmodulin; CaN, calcineurin; CRAC, calcium release-activated calcium; CREB, cAMP response element-binding protein; DAG, diacylglycerol; IL- interleukin; GM-CSF, granulocyte-macrophage colony-stimulating factor; IP_3 , inositol-3,4,5-trisphosphate; PIP_2 , phosphatidylinositol-3,4-bisphosphate; NF-AT, nuclear factor of activated T cells; NF- κ B, nuclear factor kappa-light-chain-enhancer of activated B cells; PKC, protein kinase C; RAF, rapidly accelerated fibrosarcoma; MAPKs, mitogen-activated protein kinases; MCU, mitochondrial calcium uniporter; LAT, linker for activation of T cells; Syk, Spleen Tyrosine Kinase; ZAP-70, zeta-chain associated protein kinase 70; PI3K, phosphatidylinositol 3-kinase; ERK, extracellular signal-related kinases; JNK, c-Jun N-terminal kinase; PLC γ , phospholipase C gamma, STIM1, stromal interaction molecule 1. Adapted and illustrated by C.T. Magawa in BioRender

to subsequently increase the Ca^{2+} influx speed into the mitochondria and amount via the mitochondrial calcium uniporter (MCU), even at relatively low cytosolic Ca^{2+} , potentially leading to mitochondrial Ca^{2+} overload [58].

Contrary, PregS successfully induced a significantly higher TRPM3-dependent response rate (Fig. 3D, slope, $p = 0.001$) in HC. Thus, Ca^{2+} influx amplitude via TRPM3 (Fig. 3C, $p = 0.001$) into the mitochondria was significantly reduced, with a significantly shorter $T_{1/2}$ response (Fig. 3E, $p = 0.001$) for PregS in ME/CFS compared to HC. Hence, these results show that impaired TRPM3 ion channel function may negatively impact

Ca^{2+} mobilisation into the mitochondria, potentially altering Ca^{2+} signalling downstream. A mechanistic link between TRPM3 ion channel activation, mitochondrial Ca^{2+} mobilisation and NK cell effector function is supported by evidence demonstrating that TRPM3 participates in SOCE [26], a process critically dependent on mitochondrial Ca^{2+} buffering and bioenergetic support [59–61]. Activation of TRPM3 provides a direct pathway for cytosolic Ca^{2+} influx [27, 62–64], and the resulting increase in intracellular Ca^{2+} is an absolute requirement for both cytotoxic granule exocytosis and cytokine production in NK cells [11]. Additionally, elevated cytosolic

Ca^{2+} following TRPM3 activation engages mitochondria, enhancing mitochondrial Ca^{2+} uptake [32], especially at endoplasmic reticulum-mitochondrial complexes supported by IP3R-Grp75-VDAC, meeting the energetic demands of NK cell activation [11, 65]. Thus, supporting changes in mitochondrial calcium following PregS are linked with TRPM3 ion channel activity rather than direct mitochondrial channel activation.

Importantly, this Ca^{2+} rise subsequently activates Ca^{2+} dependent enzymes, including calcineurin (CaN) and calmodulin (CaM) dependent kinases, leading to the induction of key transcriptional regulators such as nuclear factor of activated T cells (NF-AT), nuclear factor kappa B (NF- κ B) and c-Jun N-terminal kinase (JNK), which couple membrane proximal Ca^{2+} flux to effector gene expression [66–68]. In parallel, TRPM3 mediated Ca^{2+} mobilisation drives activation of phosphatidylinositol 3-kinase (PI3K) and MAPK pathways (rapidly accelerated fibrosarcoma (Raf), ERK1/2 and JNK [55, 69, 70], signalling cascades known to regulate NK cell activation, degranulation and inflammatory cytokine release following engagement of activating receptors from the natural killer group (NKG) and natural cytotoxicity receptors (NCR) families [55, 71, 72]. This shared signalling architecture highlights the versatility of TRPM3 mediated Ca^{2+} mobilisation and its potential to influence immune cell function. Evidently, PregS successfully evoked a significantly higher rate of response, accompanied by significantly higher TRPM3-dependent Ca^{2+} influx into the cytosol and mitochondria in HC compared to ME/CFS. Ononetin has been shown to effectively inhibit PregS-induced Ca^{2+} -influx and ionic currents through TRPM3 ion channels [73]. In the study, following application of ononetin alone and simultaneous application of ononetin and PregS to NK cells, there was no change in Ca^{2+} influx response, confirming that TRPM3 activity was involved in PregS-induced Ca^{2+} influx observed in NK cells (Supplementary Fig. 3). Therefore, these findings indicate that changes in Rhod-2 fluorescence indicates downstream mitochondrial Ca^{2+} influx following TRPM3 modulation (Supplementary Fig. 2). Collectively, reduced TRPM3 activity may potentially impair mitochondrial Ca^{2+} handling, attenuate MAPK and PI3K signalling, and weaken Ca^{2+} dependent effector mechanisms, consistent with the impaired TRPM3 mediated Ca^{2+} mobilisation observed in NK cells in ME/CFS. Figure 3 illustrates the proposed TRPM3-mediated signalling pathways in NK cells and potential impact in ME/CFS.

The mitochondria Ca^{2+} buffering is vital for immune cell function, however rapid Ca^{2+} uptake upon cellular stimulation leads to cytosolic Ca^{2+} rise, triggering rapid Ca^{2+} influx into the mitochondrial matrix via the mitochondrial calcium uniporter (MCU) [45, 58]. While this buffering process boosts respiration and ATP production,

excessive Ca^{2+} rise into the mitochondria contribute to dysfunction and consequently triggering cell death [74]. Impairments that lead to mitochondrial Ca^{2+} overloading contribute to cell dysfunction in neurological diseases such as Alzheimer's, Parkinson's, and Huntington's disease [74–77].

Given the vital role Ca^{2+} ions play in various important biological processes, more research focusing on exploring the potential implication of TRP channels and Ca^{2+} dysregulation to ME/CFS onset and progression are required. ME/CFS patients experience muscle weakness, sensitivities to a variety of external stimuli that may include light, sound, or specific odours [78], considering that TRP channels are involved in numerous critical physiological processes, dysregulated TRP channels may lead to Ca^{2+} imbalance mechanisms consequently generating altered signalling processes [30].

Due to the strict selection criteria applied for participant inclusion in this study, the sample size for this study was small. Therefore, many cells were analysed per participant, strategically increasing the population of cells analysed per group. Thus, $n = 554$ NK cells from $N = 10$ HC were compared with $n = 415$ NK cells from $N = 10$ ME/CFS patients for cytosolic Ca^{2+} influx examination only, with a total of six different conditions performed per group. However, GraphPad formed statistical comparison by participant, not by individual cell (nested statistical analysis), eliminating risk of pseudoreplication [40]. To mitigate potential technique limitations, methodological steps were taken, specifically, six separate wells were imaged per participant, including single dye control wells. Additionally, appropriate pharmacological agents were used including PregS and ononetin to activate and inhibit the channel, respectively. Further, FCCP was used to monitor and confirm mitochondrial uncoupling. As a thermoregulatory receptor sensitive to noxious heat at temperatures $> 37^\circ\text{C}$, the completion of experiments at RT enabled interpretation of data independent of thermoregulation. This approach ensured the drug induced response of the ion channel is characterised and aligned with previously reported electrophysiology methodology and findings for the assessment of this ion channel between groups [22, 41]. This also enabled reproducibility of previous research conducted using live cell immunofluorescent imaging and whole cell patch clamp [42, 47]. Further, due to limitations associated with the use of primary NK cells, and the finite yield obtainable from 80 mL of whole blood, it was not possible to perform all functional mitochondrial assessments concurrently. Consequently, future studies will aim to elucidate the downstream functional consequences of TRPM3 ion channel dysfunction by determining whether differences exist in mitochondrial membrane potential, ATP production, and mitochondrial reactive oxygen species

(ROS) generation between individuals with ME/CFS and HC.

In conclusion, the current study confirmed our previous findings of TRPM3 ion channel dysfunction in ME/CFS and provided new evidence demonstrating the significant impact on Ca^{2+} mobilisation into the mitochondria. Additionally, this study highlights potential downstream Ca^{2+} signalling imbalances in the mitochondria, arising from TRPM3 dysfunction. Furthermore, recent electrophysiological studies have also shown TRPM3 ion channels dysfunction in NK cells isolated from PCC patients [42]. Patients affected by ME/CFS and PCC overlap in clinical symptom presentations predominantly associated with neurological, metabolism and immunological disturbances, and a proportion of PCC patients meet the diagnostic criteria for ME/CFS. Therefore, more research focusing on finding biomarkers and providing further understanding on the pathological mechanisms involved in both illnesses is required.

Abbreviations

ATP	Adenosine Triphosphate
BMI	Body mass index
Ca^{2+}	Calcium
CaM	Calmodulin
CaN	Calcineurin
COVID-19	Coronavirus disease 2019
CNS	Central nervous system
CRAC	Calcium-release-activated calcium
CREB	cAMP response element-binding protein
DAG	Diacylglycerol
ER/SR	Endoplasmic/Sarcoplasmic Reticulum
FBC	Full blood count
FCCP	Carbonyl Cyanide 4-(trifluoromethoxy)Phenylhydrazine
F0	Minimum fluorescence
Fmax	Fluorescence at maximum peak
GM-CSF	Granulocyte-macrophage colony-stimulating factor
HC	Healthy control
MAPKs	Mitogen-activated protein kinases
MCC	Manders' colocalization coefficients
ME/CFS	Myalgic encephalomyelitis / chronic fatigue syndrome
N	Number of participants
NCNED	National centre for neuroimmunology and emerging diseases
NFAT	Nuclear factor of activated T cells
NF- κ B	Nuclear factor Kappa B
NK cells	Natural killer cells
PCC	Pearson's correlation coefficient
PCC	Post COVID-19 condition
PKC	Protein kinase C
PregS	Pregnenolone sulfate
RFI	Relative fluorescence intensity
ROI	Region of interest
SERCA	Sarco/Endoplasmic reticulum calcium-ATPase
SARS-CoV-2	Severe acute respiratory syndrome coronavirus 2
SD	Standard deviation
SF-36	36-item short-form health survey
SOCE	Store operated Ca^{2+} Entry
SPSS	Statistical package for the social sciences
Stim1	Stromal interaction molecule 1
T1	Time at minimum fluorescence intensity
T2	Time at maximum fluorescence intensity
$T_{1/2}$	Half time to peak Ca^{2+} influx
TRP	Transient receptor potential
TRPM3	Transient receptor potential melastatin 3
WHO	World Health Organisation

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12865-026-00849-1>.

Supplementary Material 1.

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Lead Contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Chandi Tabeth Magawa (chandi.magawa@griffithuni.edu.au)

Materials Availability

This study did not generate new unique reagents.

Authors' contributions

C.T.M performed all the experiments, data analysis and writing of the manuscript. S.M-G conceived the project and study coordination. C.T.M, N.E-F, K.M and S.M-G contributed to the experimental design. C.T.M, N.E-F, K.M and S.M-G contributed to critical analysis of the data. All the authors critically reviewed this manuscript and approved the final version.

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Data availability

All relevant data are present within the manuscript. The datasets from this study are not publicly available due to confidentiality agreements but could be available on reasonable request.

Declarations

Ethics approval and consent to participate

This research was conducted in accordance with Griffith University Human Research Ethics guidelines, the National Statement on Ethical Conduct in Human Research (2007, updated 2018), the Australian Code for the Responsible Conduct of Research (2018) and in accordance with Declaration of Helsinki. The human research ethics application was approved by the Griffith University Human Research Ethics Committee (GU HREC 2022/666) and all participants provided informed and written consent prior to blood collection.

Consent for publication

All participants in the study provided verbal and written informed consent for the dissemination of the anonymised data collected in the study in the form of presentations or publications.

Competing interests

The authors declare no competing interests.

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